Operations Research (OR) TD 6

Introduction to graph theory 3RD YEAR - ENSIA

An isomorphism from a simple graph G to a simple graph H is a bijection $f: V(G) \to V(H)$ such that $uv \in E(G)$ if and only if $f(u)f(v) \in E(H)$. We say "**G is-isomorphic to H**", if there is an isomorphism from G to H.

Exercice C1

Generate all possible adjacency matrices for 3-vertex graphs and draw the corresponding graphs. Identify which of these graphs are isomorphic to each other.

Exercise C2

(a) Verify that the complement of P_4 is P_4.

(b) What is the complement of K_4? of $K_{3,3}$? of C_5?

(c) Show that every self-complementary graph is connected

(d) Show that if a simple graph G is isomorphic to its complement, then the number of vertices of G has the form $4k$ or $4k$+1 for some positive integer k.

(e) Show that every self-complementary graph on $4k + 1$ vertices has a vertex of degree $2k$.

(f) Find all the simple graphs with four or five vertices that are isomorphic to their complements.

Exercice C3

(a) G is a graph with n vertices and m edges. If δ and Δ are respectively the minimum and maximum of the degrees of a graph G; show that $\delta \leq \frac{2m}{n} \leq \Delta$.

(b) If a bipartite graph $G(X, Y)$ is regular, show that $|X| = |Y|$.

(c) Prove that, if G is a simple graph with at least two vertices, then G has two or more vertices of the same degree.

Exercice C4

The line graph $L(G)$ of a simple graph G is the graph obtained by taking the edges of G as vertices, and joining two of these vertices whenever the corresponding edges of G have a vertex in common.

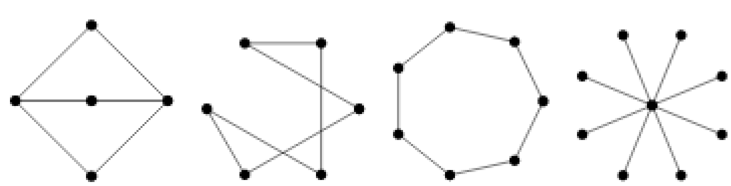
(a) Find an expression for the number of edges of $L(G)$ in terms of the degrees of the vertices of G.

(b) Show that $L(C_n)$ is isomorphic to C_n.

(c) show that $L(K_n)$ has $\frac{1}{2}n(n-1)$ vertices and is regular of degree $2n - 4$.

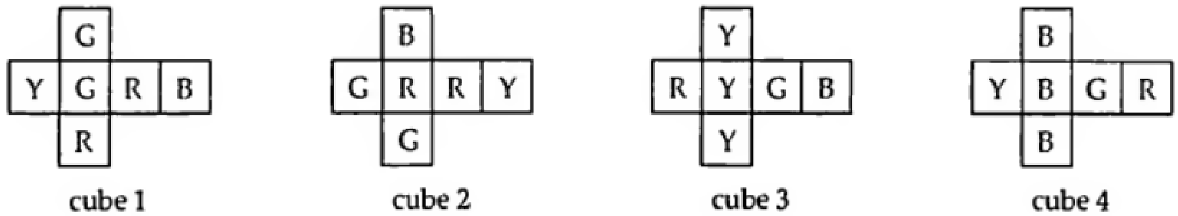
Exercice P1

Which of the graphs below are bipartite? Justify your answers.



Exercice P2

Show that there is no solution to the four cubes problem for the following set of cubes:



Exercice P3

Prove that removing opposite comer squares from an 8-by-8 checkerboard leaves a subboard that cannot be partitioned into 1-by-2 and 2-by-1 rectangles. Using the same argument, make a general statement about all bipartite graphs.

Exercice P4

Let G be a simple graph with adjacency matrix A and incidence matrix M. Prove that the degree of v_i is the ith diagonal entry in A^2 and in MM^T . What do the entries in position (i,j) of A^2 and MM^T say about G?

Exercice P5

Show that there is a gathering of five people in which there are no three people who all know each other, and no three people none of whom knows either of the other two.

Exercice P6

Prove or disprove the following statements:

(a) If G_1 and G_2 are regular graphs, then $G_1 \times G_2$ is regular.

(b) If G_1 and G_2 are bipartite graphs, then $G_1 \times G_2$ is bipartite.

Exercice P7

The following graph shows the corridors of a museum. A guard placed in a corridor can monitor two junctions placed at its ends. How many guards are needed (and how to place them) so that all intersections are monitored?

If we now place guards at intersections, assuming that a guard can monitor all corridors leading to this crossroad, how many guards are needed?